Sentiment Analysis on Movie's Reviews Using Machine Learning

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***Abstract*—This research focuses on Movies sentiment analysis through machine learning, utilizing the "IMDb ("Internet Movie Data base") dataset. Algorithms like Linear Regression, Naive Bayes, Support Vector Machines (SVM), Random Forest, Deep Learning are used. The vast amount of online data, from movie reviews to social media posts, holds valuable insights into public opinion. Sentiment analysis, also known as opinion mining, is a technique that extracts emotions and opinions from text data. This information is crucial for understanding customer preferences and market trends. As machine learning algorithms become more sophisticated and data processing techniques continue to improve, sentiment analysis will become an even more valuable tool for businesses and organizations of all stripes. By harnessing the power of emotions hidden within online data, we can unlock a deeper understanding of the people we serve, ultimately leading to better products, services, and experiences for all. This paper explores sentiment analysis in the context of movie reviews. By applying machine learning algorithms like Naive Bayes, K-Nearest Neighbors, and Random Forest, we can analyze customer reviews and gauge their sentiment towards a particular movie. The paper compares the performance of these algorithms to identify the most effective method for text analysis in sentiment analysis tasks.**

keywords— Sentiment Analysis, Machine Learning

I. INTRODUCTION

The emergence of social media sites like Twitter and Facebook has made expression more accessible to anyone. People may now express their thoughts, emotions, and evaluations on a wide range of topics on a strong platform. A data deluge—a vast volume of information in the form of postings, comments, and updates—has come from this. Opinion extraction, or sentiment analysis, uses this data to uncover hidden opinions and emotional undertones. It basically seeks to pinpoint the basic emotions portrayed in these online messages, such as happiness, sorrow, rage, and similar ones.

Sentiment analysis is not without challenges, though. Accurately assessing sentiment is complicated by slang, grammatical mistakes abbreviations, repetitious characters, the usage of regional languages, and the always changing world of emoticons. Determining the intended meaning of every word becomes a crucial undertaking.

Need of sentiment analysis on distinct levels:

1. Granularity of Opinion: Imagine analyzing a review that says "The phone has a great camera, but the battery life is terrible." A basic sentiment analysis might classify the entire review as positive because of the "great camera." However, a more nuanced analysis at the sentence level would reveal both positive and negative aspects.

2. Understanding Context: Sarcasm can be a real challenge for sentiment analysis. A comment like "This movie is fantastic, definitely not a total waste of time" needs context to understand the true sentiment. Analyzing sentiment at the phrase level can help identify sarcasm and other figures of speech that can skew the overall sentiment of a text.

3. Identifying Targets of Sentiment: A review might mention multiple aspects of a product, like screen size, battery life, and camera quality. But the sentiment might differ for each aspect. Aspect-based sentiment analysis helps pinpoint the specific features or topics that are generating positive or negative sentiment within a text.

4. Intensity of Emotion: Not all positive sentiment is created equal. A simple "good" might not convey the same enthusiasm as "absolutely love." Sentiment analysis with different levels can categorize the intensity of emotions, allowing for a more comprehensive understanding of user opinions.

Different levels and need of ratings in sentiment analysis using machine learning: -

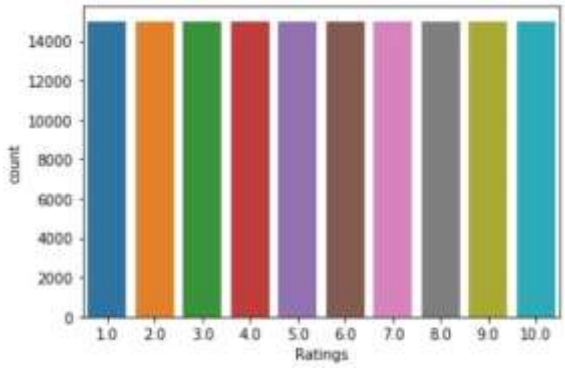
- Document-Level Sentiment Analysis: This is the most basic level, where the sentiment of an entire document (like a review, social media post, or article) is classified as positive, negative, or neutral. It provides a general sense of the overall feeling expressed.

- Sentence-Level Sentiment Analysis: This approach goes a step further by analyzing the sentiment of each individual sentence within a document. This allows for a more nuanced understanding of how sentiment might change throughout the text.

- Aspect-Based Sentiment Analysis (ABSA): This level delves deeper, focusing on identifying opinions about specific aspects of a product, service, topic, or entity mentioned in the text. It involves not only classifying sentiment but also pinpointing what exactly the sentiment is directed towards.

- Opinion Spam Detection: This specific application of sentiment analysis aims to identify fake reviews or opinions that are trying to manipulate the overall sentiment. It analyzes text for patterns and characteristics that might indicate inauthentic or biased opinions.

❖ One easy and measurable technique to measure emotion is through ratings. This makes it possible to compare goods, services, or even concepts quickly across social media platforms. To show the distribution of ratings for our dataset, we have provided a graph.



For performing sentiment analysis on any dataset, we have to go through some of the phases which are explained below: -

1. Data gathering: This initial phase involves gathering movie reviews. This can be done through various methods like scraping websites, accessing public datasets, or using APIs provided by review platforms.
2. Data Preprocessing: Raw movie reviews often contain unstructured and noisy data. This phase focuses on cleaning and preparing the data for analysis. tasks include in this phase are Lowercasing, Tokenization, Lemmatization, etc.
3. Feature Engineering: This phase involves extracting features from the preprocessed text data that will be used by the sentiment analysis model. common features include: word presence, N-grams, part-of-speech.
4. Model Selection and Training: In this phase we have to choose a sentiment analysis model based on your needs and data size.
5. Evaluate the trained model: Assess its performance on the testing set using metrics like accuracy (percentage of correctly classified samples), precision (proportion of true positives among predicted positives), recall (proportion of identified true positives), and F1 score (harmonic mean of precision and recall).

II. LITERATURE REVIEW

- Joscha et al. [1] introduce and analyze various techniques aimed at improving the performance of sentiment analysis by integrating semantic information. The authors compare techniques such as Bag of Words models and n-grams to explore the incorporation of semantic information into sentiment analysis. They note that earlier approaches to sentiment analysis often neglected the semantic associations between sentences or parts of documents. They emphasize the importance of considering semantic associations for more accurate sentiment analysis results.

- Hogenboom et al.[2] conducted research and focused on enhancing sentiment analysis using Rhetoric Structure Theory (RST), which provides a hierarchical representation at the document level. The results of the study indicated that machine learning techniques, particularly those leveraging RST and feature engineering, significantly improved the balanced accuracy and F1 score of sentiment analysis, achieving a high F1 score of 71.9%.A. Hogenboom et al. also employed feature engineering to encode binary data into random forests, thereby reducing the complexity of the original RST tree.

- Amir Hossein Yazdavar et al.[3] present a novel approach to sentiment analysis, specifically focusing on drug reviews containing numerical data. this approach involves analyzing sentences containing quantitative terms to classify them as opinionated or non-opinionated, while also determining the polarity expressed. To develop their fuzzy knowledge base, Yazdavar et al. interviewed multiple doctors from various medical centers to gather insights and domain knowledge. This approach helps ensure that the sentiment analysis model accurately captures the nuances of medical terminology and sentiment expression in drug reviews.

- The authors highlight that existing research in the field, such as work by Bhatia et al. [4], often fails to consider the presence of numerical data in reviews when determining sentiment polarity.

  Yazdavar et al. conclude that their proposed method, based on knowledge engineering using fuzzy sets, offers several advantages. They emphasize that their approach is simpler, more

efficient, and achieves a high accuracy rate, with an F1 value exceeding 72%.

- The study, authored by Liu [5], defines sentiment analysis as the domain of study that analyzes people's opinions, sentiments, appraisals, emotions, attitudes, and evaluations towards entities such as products, services, organizations, individuals, issues, events, topics, and their attributes. Liu's work establishes the broad scope of sentiment analysis, emphasizing its relevance in understanding human attitudes and perceptions across various domains.

- Nakov et al. [6], (Nakov and colleagues) proposed a sentiment analysis approach that integrates both word-level and character-level models into a unified framework. They identify three levels of sentiment processing: word, sentiment, and document. By combining different levels of analysis, Nakov et al. seek to enhance the accuracy and effectiveness of sentiment analysis models.

- The study by Hodeghatta Umesh Rao [7], presents findings from sentiment analysis conducted on Twitter messages related to Hollywood movies. Rao's work highlights the applicability of sentiment analysis techniques in social media data mining and opinion mining contexts. The analysis aims to classify tweets into positive, negative, or cognitive sentiment categories, providing insights into public perceptions and attitudes towards movie releases.

- Aston et al. [8], (Aston and colleagues) introduce algorithms leveraging perceptrons with optimized learning rates and vote perceptrons for sentiment classification of tweets in data streams. By developing efficient algorithms tailored for processing real-time data streams, Aston et al. contribute to the advancement of sentiment analysis in dynamic and fast-paced environments. By developing efficient algorithms tailored for processing real-time data streams, Aston et al. contribute to the advancement of sentiment analysis in dynamic and fast-paced environments.

- Jiang et al. [9] , this study by Jiang and collaborators propose an enhanced K-nearest neighbors (KNN) algorithm for text categorization, incorporating a combination of a one-pass clustering algorithm and KNN. The authors compare the performance of KNN and Support Vector Machine (SVM) classifiers, highlighting the superior performance of these methods over other classifiers.

- Li et al. [10], Li and co-authors conduct a comparative analysis between lexicon-based classifiers and statistical machine learning-based classifiers for sentiment analysis. They utilize the NTUSD sentiment lexicon, which comprises positive and negative sentiments, as the basis for their evaluation. Through their study, the performance of different sentiment analysis techniques, providing insights into the strengths and limitations of each approach.

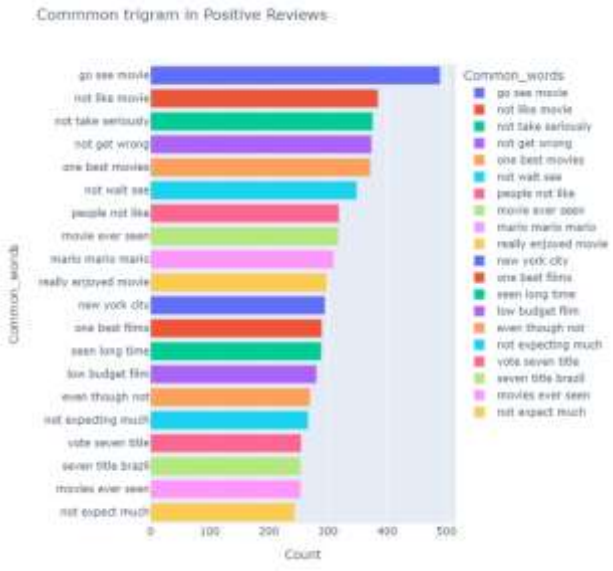
III. DATA DESCRIPTION

Sentiment analysis: The Internet Movie Database (IMDb), a popular web page where moviegoers rate and review films, is the basis of the IMDb dataset, which is an extensive collection of evaluations of movies. The objective of our initiative is to create a sentiment analysis model that classifies reviews as either positive or negative by concentrating just on two attributes: rating and review. Included in this dataset are the following important parameters:

1. Machine Learning Parameters:

Feature engineering: we can Apply the cleaned text, create instructional characteristics by using feature engineering These might consist of:

- **Bag of Words:** A key method in natural language processing (NLP) is Bag-of-Words, which converts textual data into mathematical models that artificial intelligence algorithms can interpret.
- **TF-IDF:** The TF-IDF matrices frequently serve as input characteristics for models based on machine learning, which include logistic regression, support vector machines (SVMs) or neural networks, when TF-IDF is employed in the analysis of sentiment.
- **N-gram Characteristics:** In (NLP), N-gram aspects are a crucial component of feature engineering. Consider examining bigrams or trigrams—sequences of two or three words—instead of individual words. Sentiment expressions like "laugh-out-loud" and "waste of time" can be captured by this.



2. Text Preprocessing Parameters:

Such preliminary processing workflow parameters are essentially the decisions taken to convert unprocessed text

into a format that may be used for activities like sentiment analysis, text categorization, or linguistic modeling.

- Tokenization technique: word-level tokenization, which divides the text into particular words; character-level tokenization, which divides the text into individual characters; or sub word-level tokenization, which divides words into smaller chunks like significant units or n graphs.
- Elimination of stop words: Confirm whether any stop words—words that frequently appear in texts, such as "the," "is," and "and"—were eliminated. Pre-defined stop word collections are provided by tools like as NLTK and spaCy, which can be utilized for carrying out tasks.
- Punctuation and special character handling: Explain how URLs, emojis, and other unique characters were treated throughout the preprocessing stages. Some punctuation marks can be preserved and mapped to positive or negative sentiment values.
- Stemming or lemmatization: Declare if the methods used to reduce terms to their root forms were lemmatization (which aligns words to their dictionary form) or stemming (which reduces words to their base form).
- Lowercasing: To guarantee consistency in text representation, indicate if text was changed to lowercase. It guarantees that the model handles "happy" and "Happy" in the same manner.

IV. PROPOSED METHODOLOGY

1. **Data collection:**
   Get access to the IMDb the data set, which includes ratings and reviews of films. The dataset can be obtained via academic repositories or publicly accessible datasets like those offered by Kaggle. It can also be obtained straight from IMDb.

2. **Preprocessing Data:** five steps in preprocessing data are;
   - Take out any tags within HTML, punctuation, special characters, and URLs that are present in textual content.
   - Build the input for the machine training framework, tokenize the content into separate phrases or sub words.
   - Perform Lowercasing To guarantee uniform word representation, transform every word to lowercase.
   - Remove Common Stop words that don't add much sense, such as "the", "is", and "and".
   - Use Stemming or lemmatization to standardize variants of the same word, reduce phrases to their core or root forms.

3. Extraction of Features:

- Words in a Bag (BoW): Each movie review should be shown as a vector with word counts or patterns.
- TF-IDF: Determine the TF-IDF(Term Frequency-Inverse Document Frequency) scores in order to assign a weight to each word in the reviews.
- Word insertions: Use word embedding models like Word2Vec, GloVe, or fastText, which are pre-trained, to transform words into full vector representations.

4. Model Alternatives: different models selected are;

- A basic and comprehensible model that can be used as a starting point for sentiment analysis is logistic regression.
- Another baseline model renowned for its effectiveness and simplicity is Naive Bayes.
- Support Vector Machines (SVM) is Suitable for discrete classification problems and high-dimensional data.
- Random Forest: A high-dimensional feature space and non-linear relationship handling collective learning technique.

5. Model Training: To train the model, divide the dataset into test, verification, and training sets.

- Use the relevant assessment metrics (such as accuracy and F1-score) to train the chosen artificial intelligence algorithms on the training set, and then use the validation set to adjust the hyperparameters as desired.

6. Model Evaluation:

- Using evaluation measures like accuracy, precision, recall, F1-score, and ROC-AUC, assess the trained models on the test set. To find a superior model for evaluating sentiment on the IMDb dataset, examine the results of several models.

7. Model Deployment:

- Implement the chosen model for sentiment analysis of recent film reviews in execution. Keep an eye on the model's effectiveness and update it with new data or retrain it as required.

V. DATA ANALYSIS

The movie review data from the IMDb (Internet Movie Database) dataset is examined in the data analysis part, along with the effectiveness of the machine learning models used for sentiment categorization. Machine learning methods are essential for extracting sentiment (positive, negative, or neutral) from textual data alongside. Four machine learning models: ADA Boost classifier, Logistic regression, Random forest classifier and Decision tree

classifier—will be used to examine the sentiment classification performance.

1. **ADA BOOST CLASSIFIER:**
   Algorithm explanation: Adaptive Boosting, or AdaBoost, is an ensemble learning technique that builds a powerful classifier by combining several weak classifiers. AdaBoost operates in iterations, emphasizing cases that were incorrectly categorized in earlier iterations.

   Analysis: 0.902, 0.970, 0.902 are the scores reflected for Precision score, AUC score, F1 score on training dataset for Ada Boost classifier. 0.853, 0.921, 0.853 are the scores reflected for Precision score, AUC score and F1 score on test dataset for Ada Boost classifier.

2. **LOGISTIC REGRESSION:**
   Algorithm explanation: For binary classification tasks, the linear classification approach known as logistic regression is frequently employed. In the feature space, the model learns a linear decision boundary that divides reviews with positive and negative sentiment.

   Analysis: We can obtain the Precision, Recall, AUC, F1, and Accuracy scores by using logistic regression. Thus, the assessment metrics' reflected scores of 0.896, 0.896, 0.960, 0.896, 0.896 show that the model accurately detects 89.6%, 89.6%, 96.0%, 89.6%, and 89.6% of all the dataset's favorable evaluations.

3. **RANDOM FOREST CLASSIFIER:**
   Algorithm explanation: Using a series of decision trees built during training and combining the results to arrive at the last decision, the Random Forest collaborative learning technique is used. Random Forests frequently achieve remarkable accuracy and robustness in sentiment classification problems, while being less interpretable by nature than Logistic Regression.

   Analysis: On the training dataset for the random forest classifier, the scores for Precision, AUC, and F1 are 0.999, 0.999, 0.999. Conversely, on the test dataset for the random forest classifier, the scores for Precision, AUC, and F1 are 0.854, 0.928, and 0.854.

4. **DECISION TREE CLASSIFIER:**
   Algorithm explanation: Based on a characteristic value, this method builds a structure like a tree, with each node representing an argument. Decision trees are easy to comprehend and illustrate. The model divides the data recursively according to feature values until it meets a stopping parameter.

   Analysis: 0.750, 0.819, 0.747 are the scores reflected for Precision score, AUC score, F1 score on training dataset for decision tree classifier. Whereas, 0.7192, 0.7735, 0.7160 are the scores reflected for Precision score, AUC score and F1 score on test dataset for decision tree classifier.

AdaBoost, Logistic Regression, Random Forest, and Decision Tree are the four algorithms that each have special benefits and strengths in sentiment analysis. Every algorithm had distinct benefits, meeting various demands for interpretability, performance, and scalability. Their varying performances provide important information about how well suited they are for the particular task and dataset.

VI. RESULT

The sentiment analysis of movie reviews using four machine learning methods—AdaBoost Classifier, Logistic Regression, Random Forest Classifier, and Decision Tree Classifier—provides insightful information about each algorithm's advantages and disadvantages. Using criteria such as Precision, AUC (Area Under the Curve), F1-score, and extra metrics for Logistic Regression (Recall and Accuracy), we assessed how they performed on both the training and test samples.

With good Precision (0.902), AUC (0.970), and F1 (0.902) scores on the training dataset, the AdaBoost Classifier shows promise in efficiently categorizing both favorable and adverse reviews. On the test dataset performance declined, indicating potential problems with generalization. A dependable option for sentiment analysis tasks, logistic regression shows its uniform and well-balanced effectiveness across a variety of assessment criteria on both training and test datasets. logistic regression emerges as a effective choice which gives Precision, Recall, AUC, F1, and Accuracy scores of approximately 0.896.

In random forest classifier, unusually high training set scores (almost 1.0) raised the possibility of overfitting. This notion is further supported by the notable decline in test set scores (around 0.85). The Decision Tree Classifier, on the other hand, performs moderately upon the training dataset handling Precision (0.750), AUC (0.819), and F1 (0.747) scores; but it has difficulty with generalization, as seen by lower scores (0.7192 and 0.7160) on the test dataset.

VII. CONCLUSION

In conclusion, the sentiment analysis of movie reviews using machine learning techniques provides insightful information on the efficacy of various strategies. As the AdaBoost Classifier, Logistic Regression, Random Forest Classifier, and Decision Tree Classifier were evaluated, obvious patterns concerning their functionality and task fit appeared.

TABLE 1
COMPARISON OF EVALUATION OF VARIOUS ALGORITHMS

Evaluation matrix			
Algorithms	Precision Score	AUC Score	F1 Score
ADA Boost classifier	0.902	0.970	0.902
Logistic regression	0.896	0.960	0.896
Random forest classifier	0.999	0.999	0.999
Decision tree classifier	0.750	0.819	0.747

AdaBoost Classifier's capacity to accurately categorize

sentiment was proved by its promising results on the training dataset, especially in precision, AUC, and F1 scores. Its marginally worse performance on the test dataset, however, raises the possibility of generalization issues.

The Random Forest Classifier performed exceptionally well on the training dataset, but the test dataset revealed indications of possible overfitting, as indicated by the decrease in precision and F1 scores.

Meanwhile, the Decision Tree Classifier showed mediocre performance but had trouble generalizing, suggesting that it was limited in its capacity to handle unknown data. On both training and test datasets, Logistic Regression proved to be a dependable option, exhibiting consistently balanced performance across multiple evaluation parameters. It is a good choice for sentiment analysis jobs because of its robustness and interpretability.

VIII. FUTURE SCOPE

The project's ongoing efforts will concentrate on improving machine learning techniques for enhanced sentiment analysis accuracy. Convolutional neural networks (CNNs) and recurrent neural networks (RNNs) are examples of deep learning architectures that we may use to automatically extract features from reviews and possibly increase accuracy. Furthermore, obtaining and compiling very big movie review datasets—possibly from sources other than IMDb—may entail paying license fees or requiring a sizable amount of processing power. Thus, continuing refinement and future breakthroughs in novel approaches can reduce processing prices. Support trust and transparency by incorporating XAI approaches to comprehend how machine learning models arrive at sentiment classifications on movie evaluations. This lets you find any biases or constraints in the way the model makes decisions. Design recommendation systems regarding movie sites that make suggestions for films that people might like based on their past feelings about the films. Sentiment analysis can enhance user participation and assist tailor movie recommendations.

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